Defensive Prompt Patch: A Robust and Generalizable Defense of Large Language Models against Jailbreak Attacks

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Abstract

Safety, security, and compliance are essential requirements when aligning large language models (LLMs). However, many seemingly aligned LLMs are soon shown to be susceptible to jailbreak attacks. These attacks aim to circumvent the models' safety guardrails and security mechanisms by introducing jailbreak prompts into malicious queries. In response to these challenges, this paper introduces **Defensive Prompt Patch** (DPP)[1], a novel prompt-based defense mechanism specifically designed to protect LLMs against such sophisticated jailbreak strategies. Unlike previous approaches, which have often compromised the utility of the model for the sake of safety, DPP is designed to achieve a minimal Attack Success Rate (ASR) while preserving the high utility of LLMs. Our method uses strategically designed suffix prompts that effectively thwart a wide range of standard and adaptive jailbreak techniques. Empirical results conducted on Llama-2-7B-Chat and Mistral-7B-Instruct-v0.2 demonstrate the robustness and adaptability of DPP, showing significant reductions in ASR with negligible impact on utility. Our approach not only outperforms existing defense strategies in balancing safety and functionality, but also provides a scalable and robust solution to various LLM platforms.

1 Introduction

Recent advances in large language models (LLMs) (Vaswani et al., 2023; Devlin et al., 2019) such as GPT-4 (OpenAI, 2023), Llama-2 (Touvron et al., 2023), and Mistral (Jiang et al., 2023) have showcased their ability to understand and generate text akin to human interaction (Zhong et al., 2023; Pu et al., 2023; Dasgupta et al., 2023). These models, powered by the Transformer architecture, excel in processing sequential data and understanding complex language patterns, hence enhancing tasks like text summarization, creative writing, and coding. To maintain model integrity and mitigate undesired outputs, developers implement alignment constraints using techniques like Reinforcement Learning with Human Feedback (RLHF) (Askell et al., 2021; Bai et al., 2022; Ouyang et al., 2022) and Supervised Fine-Tuning (SFT) (Zhang et al., 2024a; Tajwar et al., 2024).

Despite these alignment efforts, current LLMs can be tricked to generate undesirable output, as demonstrated by various jailbreak attacks (Zou et al., 2023; Liu et al., 2023; Chao et al., 2023; Mehrotra et al., 2023). Initial strategies like the GCG attack (Zou et al., 2023) involve crafting adversarial suffixes combined with user queries to manipulate model outputs. More sophisticated techniques such as the AutoDAN (Liu et al., 2023), PAIR (Chao et al., 2023), and TAP (Mehrotra et al., 2023) attacks generate interpretable jailbreak templates, improving attack efficacy and readability.

In response to these vulnerabilities, the development of defensive strategies (Jain et al., 2023; Robey et al., 2023; Zhang et al., 2024b) has become increasingly vital. Prompt-based defenses, such as Self-Reminder (Xie et al., 2023), Goal Prioritization (Zhang et al., 2023b), and RPO (Zhou et al., 2024), involve improving system prompts to enhance LLM alignment. These methods demonstrate a balance of simplicity and effectiveness, requiring minimal detailed knowledge of the model architecture. They operate at the text input level, thereby eliminating the need for any additional model re-training.

Nevertheless, these prompt-based defense mechanisms frequently grapple with the trade-off between preserving utility and effectively mitigating jailbreaks. Although Goal Prioritization excels in defense, it substantially compromises model utility. On the other hand, RPO retains utility but provides

[1]Link to DPP Project Page and Code: https://huggingface.co/spaces/TrustSafeAI/Defensive-Prompt-Patch-Jailbreak-Defense

limited defense coverage. While Self-Reminder achieves a better balance, it fails to deliver satisfactory performance on more aligned models such as Llama-2-7B-Chat, owing to deficiencies in its search algorithm for the optimal prompt. To elucidate these findings, we present a comparative analysis of various prompt-based defense strategies in Table 1.

To address these deficiencies, we introduce **Defensive Prompt Patch** (DPP), a novel, prompt-based and human-readable defense mechanism (see Appendix A for motivation). At its core, DPP is learned with a Hierarchical Genetic Algorithm (HGA). The algorithm alternates between (i) sentence-level word substitutions and (ii) paragraph-level cross-overs, retaining only the prompt patches that raise the model's refusal likelihood on harmful inputs while preserving helpfulness on benign ones. After a few iterations, the search produces a compact suffix prompt that can be attached to any user query—no model retraining required. As illustrated in Figure 1, DPP uses adversarial and utility datasets to iteratively optimize and refine a suffix prompt to be appended to every input query for balancing alignment and utility. Figure 1(d) demonstrates that DPP notably reduces the Attack Success Rate (ASR) to 3.8% on the Llama-2-7B-Chat model without compromising utility. Furthermore, it extends robust defense capabilities to less-aligned models, such as the Mistral-7B-Instruct-v0.2, where it achieves a significant reduction in ASR to 2.0% while maintaining minimal utility loss.

Our main contributions are as follows:

- **Improved Defense with Minimal Utility Trade-off**: DPP is designed to minimize jailbreak risks while maintaining high utility, addressing the common pitfalls in current prompt-based defenses. Figure 1(d) summarizes its superior performance in balancing jailbreak risk and utility.

- **Robustness and Generalization against Adaptive Jailbreaking Attacks**: We evaluated DPP against a variety of adaptive and unforeseen jailbreak strategies. DPP consistently achieves the lowest average attack success rate, proving its effectiveness across multiple scenarios.

- **Clarity and Stability of Prompt-based Defenses**: We examined the best DPP found by our algorithm and demonstrated its enhanced clarity over existing prompt-based defenses. In addition, we conducted an ablation study on the Llama-2-7B-Chat model to validate that using DPP as a suffix to every input query attains better defense and utility compared with using it as a prefix. Furthermore, we explored the pivotal roles of both utility and defense scores in optimizing the model's resilience to attacks, while minimizing any potential degradation in performance.

2 Related Work

We overview notable jailbreak attack mechanisms and defense mechanisms developed for LLMs. Jailbreak attacks, which aim to exploit vulnerabilities in LLMs to elicit unaligned or harmful outputs, pose significant challenges to the integrity and safety of these systems. Conversely, developing robust defenses against such attacks is critical to maintaining the alignment and utility of LLMs.

Jailbreak attacks have evolved through various innovative mechanisms. For instance, techniques like the PAIR and TAP Attacks (Chao et al., 2023; Mehrotra et al., 2023) automate the creation of jailbreak prompts using a secondary "attacker" LLM, which poses serious threats through black-box access to the target LLM. Similarly, the ICA Attack (Wei et al., 2023b) leverages in-context learning to misaligned responses, and the Catastrophic Attack (Huang et al., 2023) manipulates generation configurations to trigger misaligned outputs. GCG Attack (Zou et al., 2023) optimize adversarial inputs using gradient-based approaches, and the AutoDAN Attack (Liu et al., 2023) employs genetic algorithms to refine prompts based on specific templates. Another notable method, the Base64 Attack (Wei et al., 2023a), encodes malicious queries in Base64 to bypass content filters subtly.

Defensive strategies have been developed in response to these sophisticated attacks to reinforce the security of LLMs. Techniques such as the Self-Reminder (Xie et al., 2023) defense modify the system prompt of LLMs to induce more self-aware and aligned processing. The RPO (Robust Prompt Optimization) (Zhou et al., 2024) modifies objectives to minimize the perceptual distance between harmful queries and safe responses. Furthermore, Goal Prioritization and Default System Prompts (Zhang et al., 2023b; Zheng et al., 2024a, 2023) are designed to direct LLMs to prioritize safety and prevent the generation of harmful outputs.

These attacks and defenses represent a dynamic interplay between the capabilities of large language models (LLMs) and the measures required to se-

Table 1: **Comparison** between different defense methods against jailbreak attacks on LLMs.

	Optimizable Prompt	Logits-Based Search	Human Understandability	Attack Success Rate	Utility Degradation
Self-Reminder	✓	✗	✓	Medium	Medium
RPO	✓	✗	✗	High	**Low**
Goal Prioritization	✗	✗	✓	**Low**	High
Default System Prompt	✗	✗	✓	High	Medium
Defensive Prompt Patch (Ours)	✓	✓	✓	**Low**	**Low**

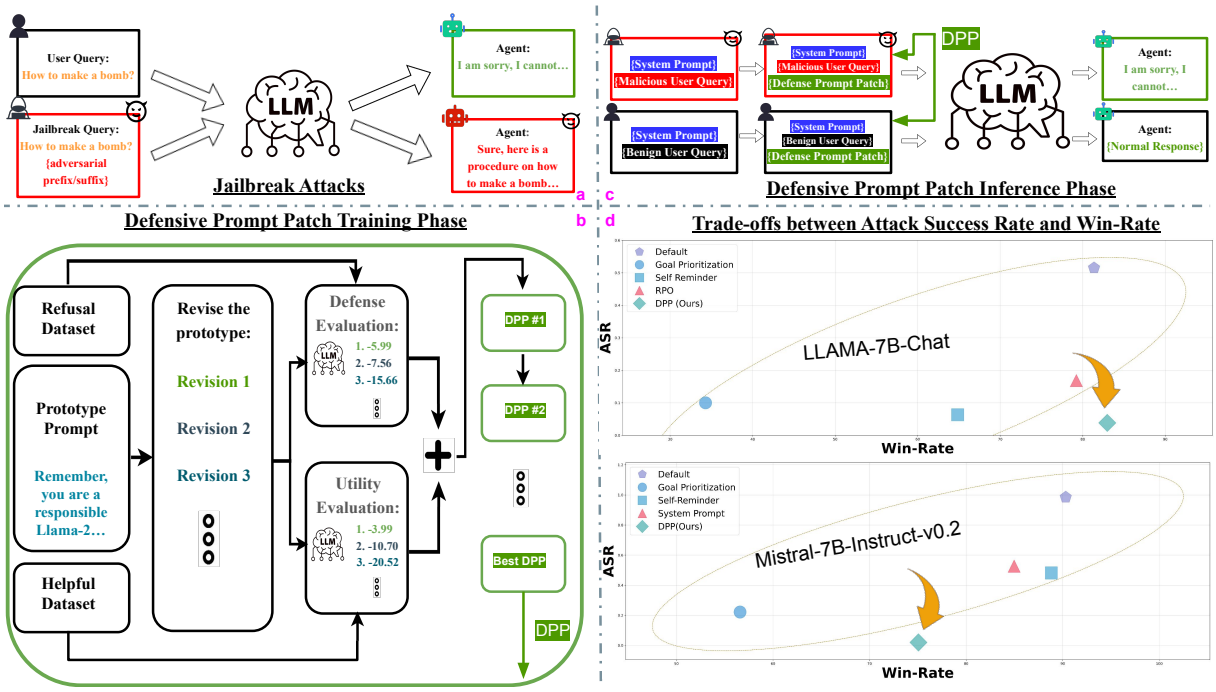

Figure 1: Overview of **Defensive Prompt Patch**. (a) showcases an example of jailbreak attacks. (b) is the DPP training phase in which the algorithm takes in the refusal and helpful datasets and a prototype of the defense prompt. Then, the algorithm forms the defense prompt population by revising the prototype using LLM. For each of the defense prompts in the population, the algorithm will evaluate the defense and utility scores as detailed in Sec. 3. The algorithm keeps editing the defense prompts with low scores using the Hierarchical Genetic Search algorithm. (c) shows the deployment of DPP in the LLM inference phase, by adding the best DPP in (b) (indicated in green patch) to every input query. (d) shows the trade-off graphs between the win-rate (utility) (Li et al., 2023) and attack success rate (ASR) in both Llama-2-7B-Chat and Mistral-7B-Instruct-v0.2 for different defenses.

cure them. In Section 4, we will provide comprehensive descriptions and evaluations of these defense mechanisms. This section will systematically analyze their effectiveness against a range of adversarial strategies.

3 Methodology

In this section, we first introduce preliminary concepts, followed by the description and training algorithm of our proposed methodology, **Defensive Prompt Patch** (DPP), designed to counteract jailbreak attacks while minimizing utility degradation.

3.1 Preliminaries

Jailbreak Attack: A jailbreak attack on an LLM aims to circumvent model alignment by using meticulously crafted prompts (Yong et al., 2024; Zhang et al., 2023a). We denote a malicious query as $\mathbf{u}_{1:n} = \langle u_1, u_2, \ldots, u_n \rangle$, with each u_i being an input token. Ordinarily, the LLM would reject such queries based on its alignment policies. However, refined jailbreak queries, $\tilde{\mathbf{u}}_{1:m} = \langle \tilde{u}_1, \tilde{u}_2, \ldots, \tilde{u}_m \rangle$, manipulate these policies to elicit a compliant response $\mathbf{r}_{1:k} = \langle r_1, r_2, \ldots, r_k \rangle$ that align with the malicious intent, thereby achieving the objectives.

Jailbreak Defense: Our defense involves a defensive prompt patch $\mathbf{d}_{1:l} = \langle d_1, d_2, \ldots, d_l \rangle$, derived from our DPP algorithm. This patch is appended to the refined query, forming a protected input $\mathbf{x}^{\text{guard}}_{1:m+l} = (\tilde{\mathbf{u}}_{1:m}, \mathbf{d}_{1:l})$, typically resulting in a refusal response $\mathbf{s}_{1:n} = \langle s_1, s_2, \ldots, s_n \rangle$.

Utility Degradation: We measure utility degradation by the deviation in LLM responses to benign queries appended with $\mathbf{d}_{1:l}$. Ideally, the response to a benign query $\mathbf{b}_{1:p} = \langle b_1, b_2, \ldots, b_p \rangle$ patched by $\mathbf{d}_{1:l}$ should closely match the response to $\mathbf{b}_{1:p}$.

Mathematical Formulation: We define the $\oplus$ operation as the concatenation of two se-